

## Assessment 4

### Server Security and Automation

#### Section A: Scenario-Based Concept Application

1. Explain the operational factors that justify selecting a specific Linux distribution for a startup's internal tool hosting environment.

ANS :

❖ When choosing a Linux distribution for hosting internal tools, consider:

- **Stability:** Use distributions like Ubuntu server or Red Hat Enterprise Linux for reliable operation.
- **Security Updates:** Regular security patches reduce vulnerabilities.
- **Community & Support:** Large communities provide documentation and troubleshooting help.
- **Package Availability:** Easy installation of Apache, MySQL, PHP, Docker, etc.
- **Cost:** Ubuntu is free, while RHEL offers paid enterprise support.
- **Hardware Compatibility:** Must support existing server hardware.

2. Explain the implementation of SSH Key-based Authentication to satisfy a security requirement where password logins must be disabled for developer access.

ANS :

❖ Implementation:

- 1) Generate key pair:
  - `ssh-keygen -t rsa -b 4096`
- 2) Copy public key to server:
  - `ssh-copy-id user@server-ip`
- 3) Verify login:
  - `ssh user@server-ip`
- 4) Edit SSH configuration:
  - `sudo nano /etc/ssh/sshd_config`
- 5) Restart SSH:
  - `sudo systemctl restart sshd`

❖ **Benefits:**

- Stronger security
- Prevents brute-force password attacks
- Convenient for developers

3. Explain the configuration of Linux Permissions and Ownership for the /var/www directory to support a multi-user environment.

ANS :

1. Create group:
  - `sudo groupadd webadmins`
2. Add users:
  - `Sudo usermod -aG webadmins user1`  
`sudo usermod -aG webadmins user2`
3. Change ownership:
  - `sudo chown -R root:webadmins /var/www`
4. Set permissions:
  - `sudo chmod -R 775 /var/www`

❖ **Result:**

- Owner: Full control
- Group: Read/Write/Execute
- Others: Read/Execute only

➤ This allows multiple administrators to manage website files safely.

4. Explain how Logical Volume Management (LVM) facilitates dynamic storage expansion to prevent service downtime when disk space is depleted.

ANS :

➤ LVM allows storage to grow without downtime.

Steps:

- 1) Add new disk.
- 2) Create Physical Volume:
  - `pvcreate /dev/sdb`
- 3) Extend Volume Group:
  - `vgextend vgdata /dev/sdb`
- 4) Extend Logical Volume:
  - `lvextend -L +20G /dev/vgdata/lvweb`
- 5) Resize filesystem:
  - `resize2fs /dev/vgdata/lvweb`

- ❖ Benefits:
  - Online expansion
  - No service interruption
  - Flexible storage management

5. Explain the troubleshooting process for diagnosing and resolving service blocks caused by SELinux security policies when a website becomes inaccessible.

ANS :

- 1) Check SELinux status:
    - `getenforce`
  - 2) View denied operations:
    - `sudo ausearch -m AVC`
  - 3) Check audit logs:
    - `sudo cat /var/log/audit/audit.log`
  - 4) Generate explanations:
    - `sudo sealert -a /var/log/audit/audit.log`
  - 5) Apply proper context:
    - `sudo restorecon -Rv /var/www/html`
  - 6) Allow required port:
    - `sudo semanage port -a -t http_port_t -p tcp 8080`
  - 7) Verify:
    - `systemctl restart httpd`
6. Explain the safe configuration of Cron Jobs to ensure critical daily backup tasks execute automatically at a scheduled time.

ANS :

1. Create backup script:
    - `tar -czf /backup/www_$(date +%F).tar.gz /var/www`
  2. Make executable:
    - `chmod +x backup.sh`
  3. Edit cron:
    - `crontab -e`
  4. Add:
    - `0 2 * * * /home/user/backup.sh >> /var/log/backup.log 2>&1`
- ❖ Benefits:
    - Automatic backups
    - Logging enabled
    - Minimal human intervention

## **Section B: Practical Application**

1. System Identity: Set the system hostname to startup-web-01 and verify the change via CLI.

ANS :

- Set hostname:
  - `sudo hostnamectl set-hostname startup-web-01`
- Verify:
  - `hostnamectl`  
`hostname`

2. Network Setup: Configure a static IP address on the primary interface and restart the networking service.

ANS :

❖ Command :

```
ip addr
nmcli con show
sudo nmcli con mod ens33 ipv4.addresses 192.168.1.100/24
sudo nmcli con mod ens33 ipv4.gateway 192.168.1.1
sudo nmcli con mod ens33 ipv4.method manual
sudo systemctl restart NetworkManager
ip addr
```

3. Secure Access: Enable the SSH service and ensure it starts automatically on system boot.

ANS :

- Install SSH:
  - `sudo apt install openssh-server`
- Enable at boot:
  - `sudo systemctl enable ssh`  
`sudo systemctl start ssh`
- Verify:
  - `sudo systemctl status ssh`

4. Access Control: Create a group named webadmins and a user dev-ops with restricted sudo privileges.

ANS :

1. Create group:
    - sudo groupadd webadmins
  2. Create user:
    - sudo useradd -m dev-ops
    - sudo passwd dev-ops
  3. Restricted sudo:
    - sudo visudo
  4. Add:
    - dev-ops ALL=(ALL) /usr/bin/systemctl restart apache2
5. Web Hosting: Install the Apache service and deploy a custom index.html file to /var/www/html.

ANS :

1. Install Apache:
  2. RHEL/CentOS:
    - sudo yum install httpd -y
  3. Create webpage:
    - echo "<h1>CloudScale Startup Server</h1>" | sudo tee /var/www/html/index.html
  4. Start service:
    - sudo systemctl enable apache2
    - sudo systemctl start apache2
6. Hardening: Configure the system firewall to permit traffic only on ports 22 and 80.

ANS :

#### ❖ RHEL/CentOS (Firewalld)

- sudo firewall-cmd --permanent --add-port=22/tcp
- sudo firewall-cmd --permanent --add-port=80/tcp
- sudo firewall-cmd --reload

7. Automation: Script a cronjob that executes at 02:00 daily to compress /var/www into a .tar.gz archive.

ANS :

- 1) Create script:
  - nano /usr/local/bin/www\_backup.sh
- 2) Content:
  - tar -czf /backup/www\_\$(date +%F).tar.gz /var/www
- 3) Permissions:
  - chmod +x /usr/local/bin/www\_backup.sh
- 4) Cron entry:
  - crontab -e
- 5) Add:
  - 0 2 \* \* \* /usr/local/bin/www\_backup.sh

### **Section C: Integrated Case Study**

Scenario: "CloudScale," a boutique hosting provider, recently performed a kernel update on their primary web server. Upon reboot, the web application is unreachable (502 Gateway Error), and administrators report that SSH login attempts take over 60 seconds to respond. The server is live, but the business is losing revenue due to the downtime and unresponsive management interface.

1. Diagnosis: Which specific log files and system commands would you use to identify why the Apache service failed to start?

ANS :

- 1) Check status:
  - systemctl status apache2
- 2) View logs:
  - journalctl -xe
- 3) RHEL:
  - cat /var/log/httpd/error\_log
- 4) Check configuration:
  - apachectl configtest

2. Connectivity: Analyze why the SSH login is experiencing high latency and suggest a fix related to DNS or authentication settings.

ANS :

❖ **DNS Reverse Lookup**

1. Edit:
  - `sudo nano /etc/ssh/sshd_config`
2. Add:
  - `UseDNS no`
3. **GSSAPI Authentication Delay**
  - `GSSAPIAuthentication no`
4. Restart SSH:
  - `sudo systemctl restart sshd`

❖ **Result:**

- SSH response becomes immediate
- No unnecessary DNS lookup delays

3. Recovery: Outline the step-by-step CLI process to manually restart the failed services and verify their operational status.

ANS :

1. Check services:
  - `systemctl status apache2`  
`systemctl status sshd`
2. Restart:
  - `sudo systemctl restart apache2`  
`sudo systemctl restart sshd`
3. Verify listening ports:
  - `ss -tulpn`
4. Verify website:
  - `curl http://localhost`
5. Verify SSH:
  - `ssh localhost`

4. Preventive Action: Propose a configuration change to ensure the web server automatically attempts to restart if it encounters a runtime failure.

ANS :

1. Create systemd override:
  - `sudo systemctl edit apache2`
2. Add:
  - `[Service]`  
`Restart=always`  
`RestartSec=5`
3. Reload:
  - `sudo systemctl daemon-reload`  
`sudo systemctl restart apache2`

✓ Now Apache automatically restarts after failures.

5. Security: Justify the use of SSH Key-based authentication over passwords to prevent brute-force attacks during this vulnerable state.

ANS :

### **SSH Keys**

Very difficult to crack

Uses public/private cryptography

Automated secure access

No password transmission

### **Passwords**

Vulnerable to brute force

Relies on password strength

Users often reuse passwords

Password attacks possible